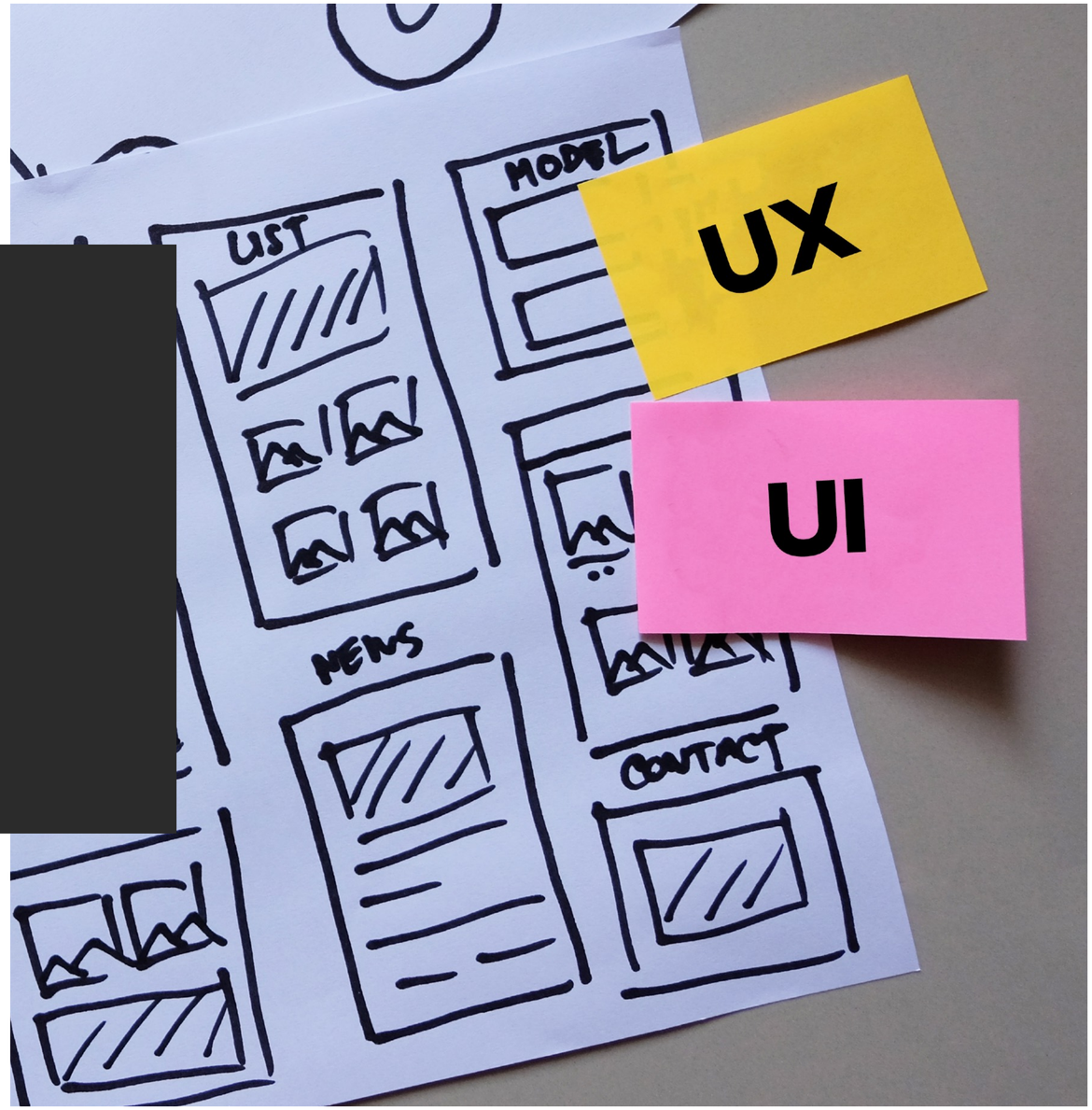
User Experience Design

Ing. Lisa-Maria Freller, MA

# Requirements & Low-Fi prototyping

# Overview

| | |
|---|---|
| 13/10 | Requirements & Low-Fi PT |
| 15/10 | High-Fi PT |
| 30/10 | Check in (online) / Tests |
| 20/11 | Check in / Tests |
| 11/12 | Check in / Tests |





## UX/UI Designer · FE-Developer from industrial background

**Research assistant** · until 2023 · industry experience

**Part-time lecturer** · since 2023

**UX/UI Designer** @ inola · since 2024

**HTL Leonding**

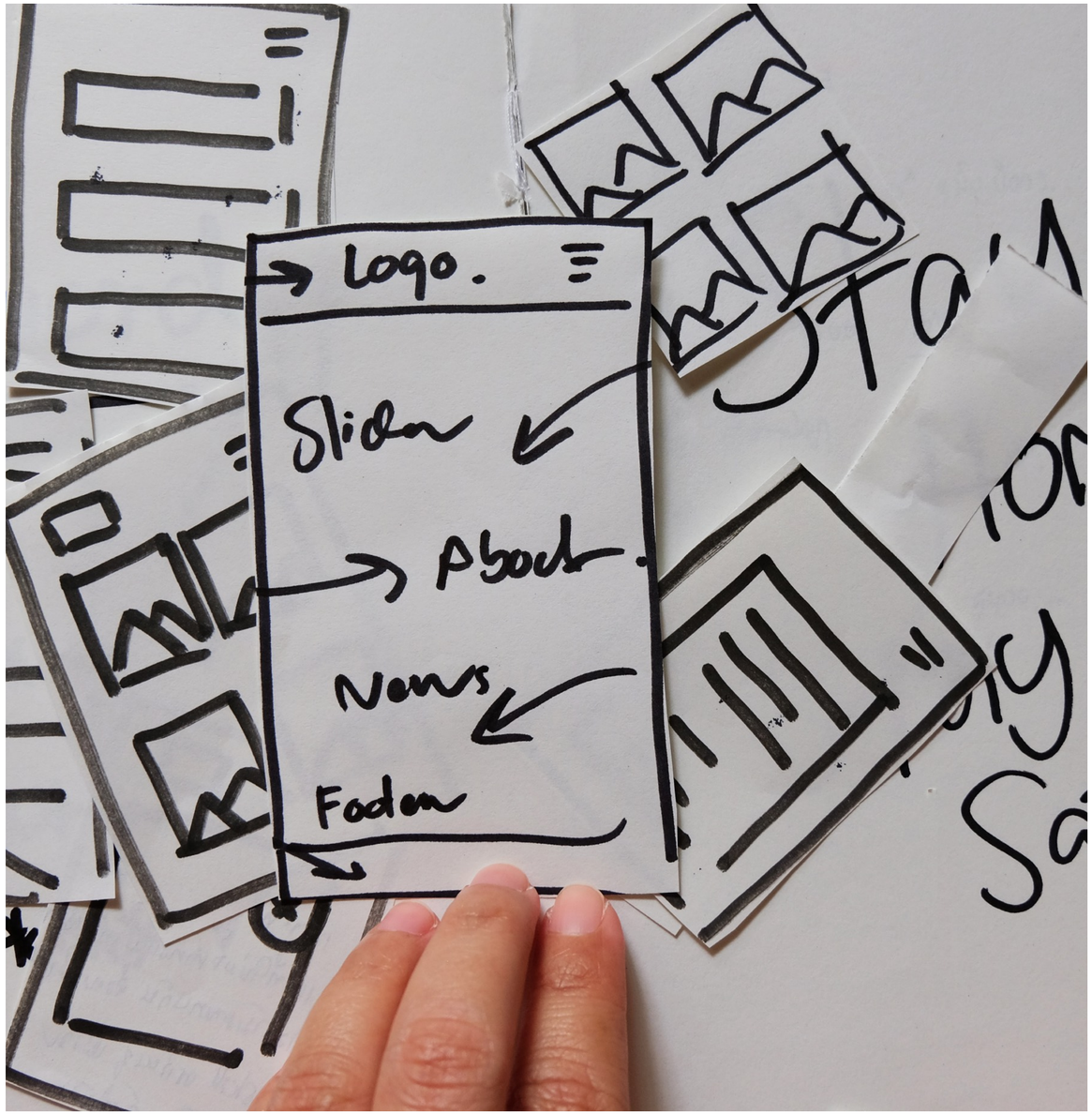
Computer Science & Media Technology

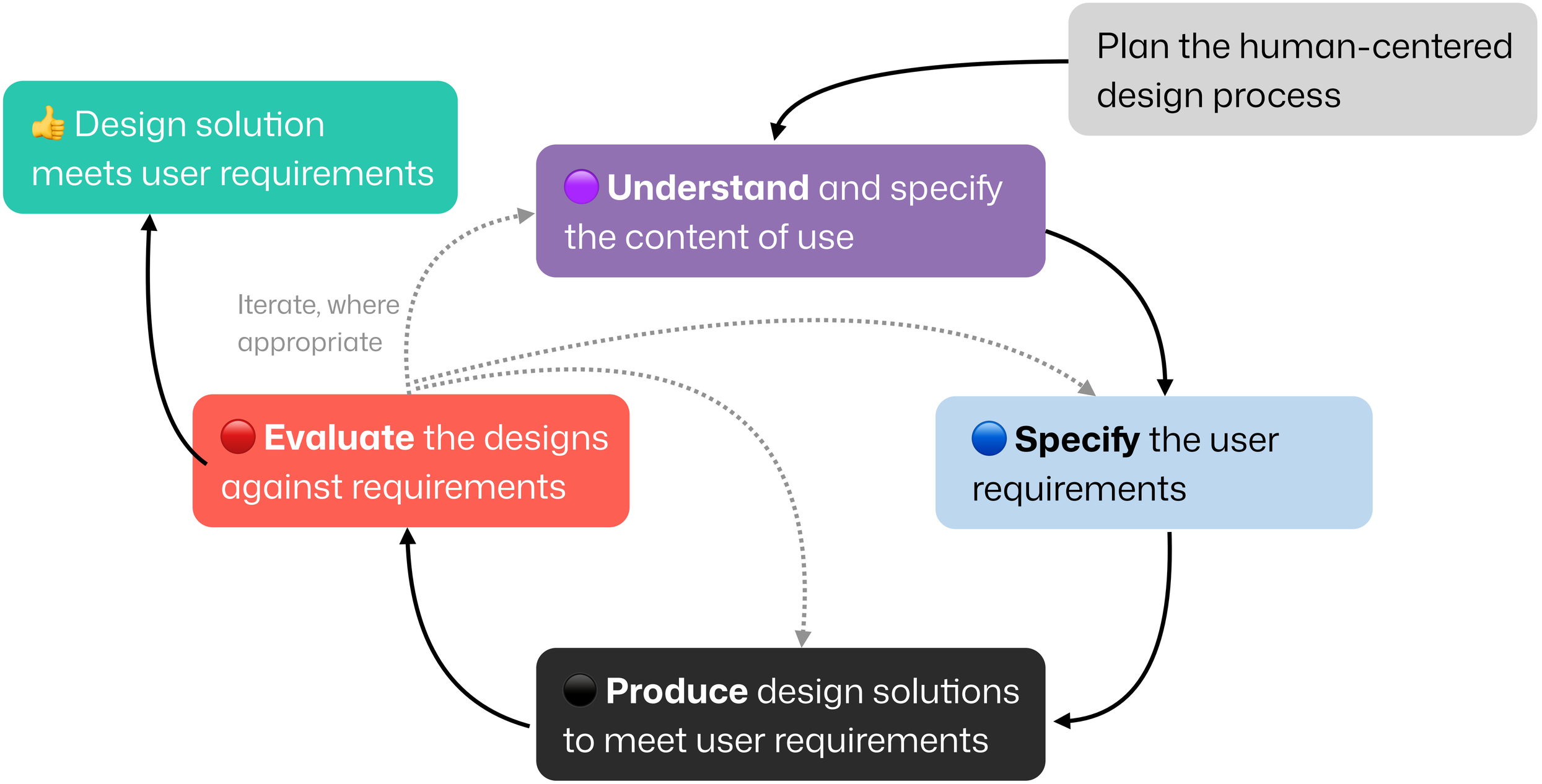
**BA & MA @ Hagenberg**

Communication & Knowledge Media

# Today

- **Requirements**
  - Understand problem and define what users need
- **Prototyping**
  - Explore and visualise how the design could meet those needs




Plan the human-centered design process
👍 Design solution meets user requirements
🟣 Understand and specify the content of use
Iterate, where appropriate
🔴 Evaluate the designs against requirements
🔵 Specify the user requirements
⚫ Produce design solutions to meet user requirements

| | |
|---|---|
| 🟣 **Persona** | Represents research based user group, their attitudes, goals & behaviors |
| 🟣 **Context Scenario** | Story describing how persona interacts to achieve a goal within context |
| 🟣 **Journey Map** | Visualises needs & pain points across phases & touchpoints |
| 🟣 **Empathy map** | Organises what users say, think, do, and feel |
| 🔵 **User story** | Expresses user needs and goal, used by agile teams |
| ⚫ **Wireframe** | Defines structural layout and content of an interface |
| ⚫ **User Flow** | Visualises steps and decisions users take to accomplish a goal |
| ⚫ **Storyboard** | Visualizes a scenario through a sequence of images |
| 🔴 **User Testing** | Evaluates whether users can achieve goals effectively and efficiently |

# Requirements

# Persona

## Prototypical, research-based representation of target user

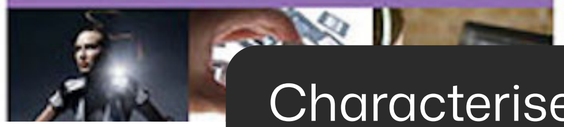
- ✓ Fictional representations and generalizations of clusters of target users with **similar attitudes, goals, and behaviours** in relation to product
- ✓ **Human-like** snapshots of relevant and meaningful similarity
- ✓ **Guides design decisions** by keeping user needs in focus



PERSONA PROFILE



"I want to get as many of my images out there to make a name for myself in this creative industry."



## Jade The Emerging Creative

Age: 29
Occupation: Commercial Photographer
Location: Chicago, IL

With three years of experience in commercial photography, most experts would consider Jade a fresh face in the industry. She's tired of the label and wants her work to be more recognized.

Jade's been building her portfolio with unique and edgy photoshoots with marketing appeal that she sends to clients and Corbis Images. She does it all on the creative side, from creating the concept, working with the models, and editing photos. Jade enjoys the post production work the most, as she is able to utilize the latest technology, from touching up subjects to incorporating computer generated elements to her photos. While Jade enjoys the aristic aspect, she dislikes dealing with administrative tasks such as model releases. Jade wants her clients to trust that her creative photography can help sell their product, but that trust comes with building a reputable name in the industry.

## Jade's Questions

## Jade's Goals

- I want to make a name for myself in the creative industry. I am looking to develop my prestige and reputation.
- There are many steps to a successful photo shoot and I would rather not deal with the complicated administrative tasks. I want something that can streamline these tasks into my workflow.
- I enjoy thinking outside of the box and having someone to generate ideas with.

| | | LOW | MID | HIGH |
|---|---|---|---|---|
| NEEDS | Speed | | | |
| | Adaptability | | | |
| | Editor | | | |
| TASKS | Upload | | | |
| | Input Info | | | |
| | Royalties | | | |

Characterise different user groups to better understand **needs, goals, behaviours, and preferences**.

By looking at the product from the persona perspective, we can build **empathy for users** and understand their experiences.

## UX persona

**Question** • Who will use the product and what do they need to accomplish?

**Example** • done by project team redesigning the bill-pay experience for customers with checking account

**Scope** • specific to product, **granular,** information built into personas positively influences user experience of new feature

## Marketing Persona

**Question** • Who are we trying to reach and why would they choose us?

**Example** • represent customers for various banking products (savings / checking account)

**Scope** • focus often **broad,** which makes them less specific to individual product

✓ When describing ux personas, we focus **only** on user characteristics that **directly influence the design** of an solution

X Often, people want to (re)use personas created for a different purpose



# Context scenario

## Describe product requirements from the user perspective through narrative descriptions of tasks users will undertake to achieve their goals

- ✓ Describe the **experience**, not interface details
- ✓ Focus on **what happens** and avoid adding detail of how it happens
- ✓ Often used for **ideation** of new design ideas or testing

persona

**Debbie** is going on a business trip. **She needs to book a hotel room** that's affordable and has good reviews.

intention

context

**It is Monday morning** and **Debbie is at work.**

Debbie **browses the site** to find a hotel for her upcoming trip. She looks closely at the various hotels to find one that meets her needs. She considers price and user ratings heavily as she shops.

action

resolution

Debbie **selects a hotel and books a room.**

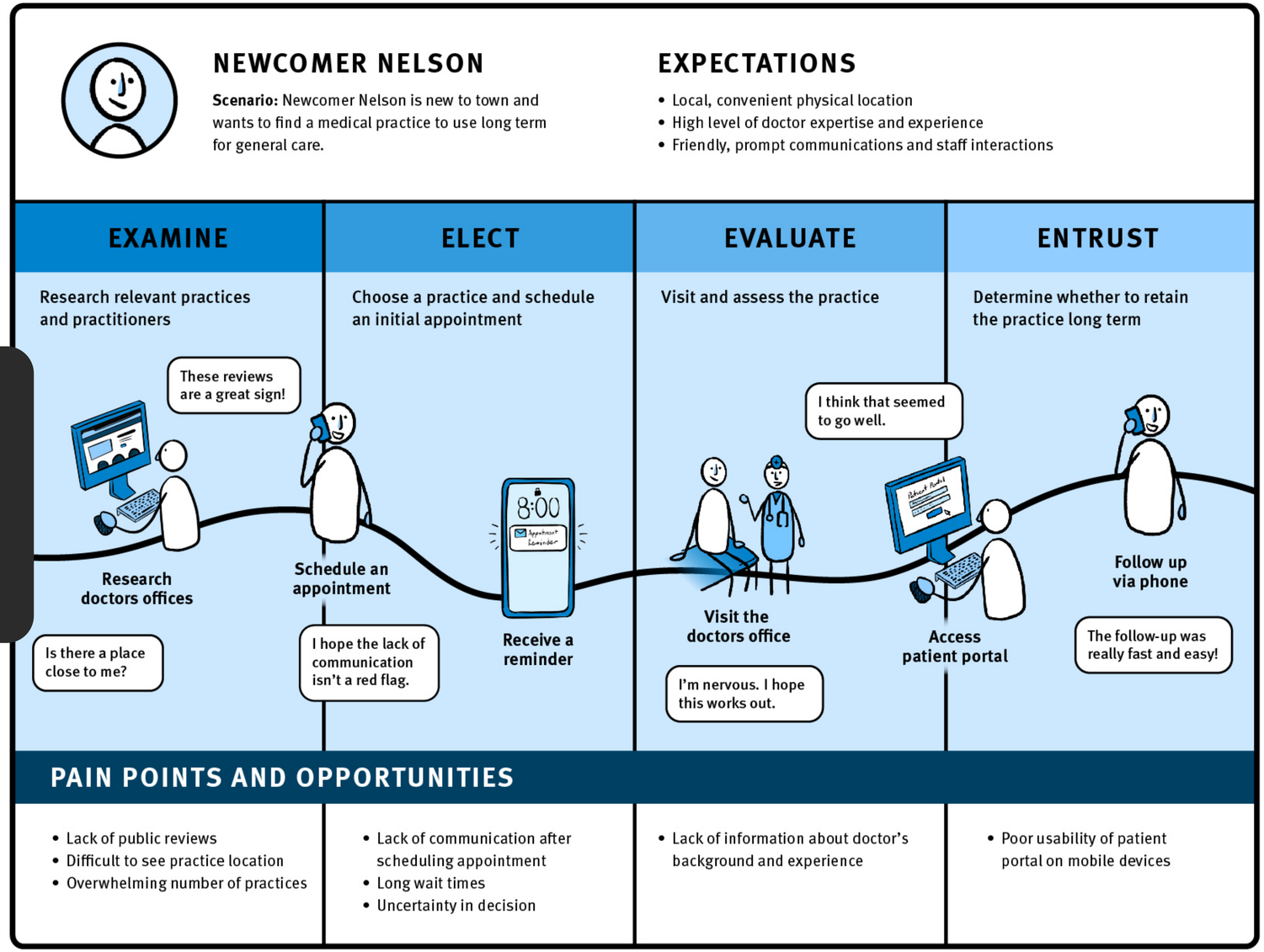
# User journey: Become new patient




A scenario-based **sequence of steps** a user takes to achieve a goal with product(s) **over time**

# Journey Map

Visual representation of journey showing **actions, thoughts** & **feelings** of users

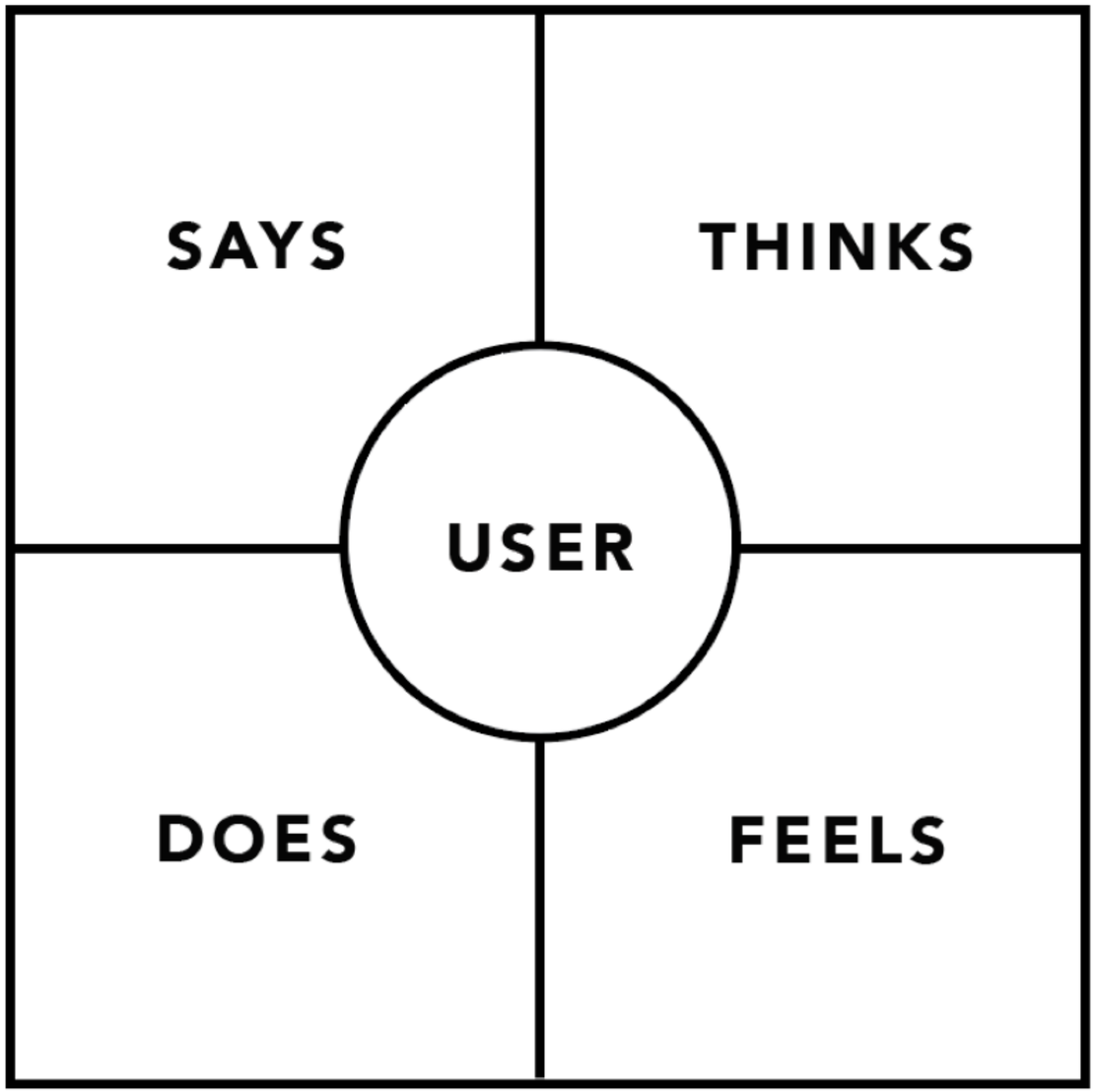



https://www.nngroup.com/articles/user-journeys-vs-user-flows/

# Empathy Map

## Used to articulate what we know about a particular type of user

It externalises knowledge about users in order to create a shared understanding of user needs, and aid in decision making



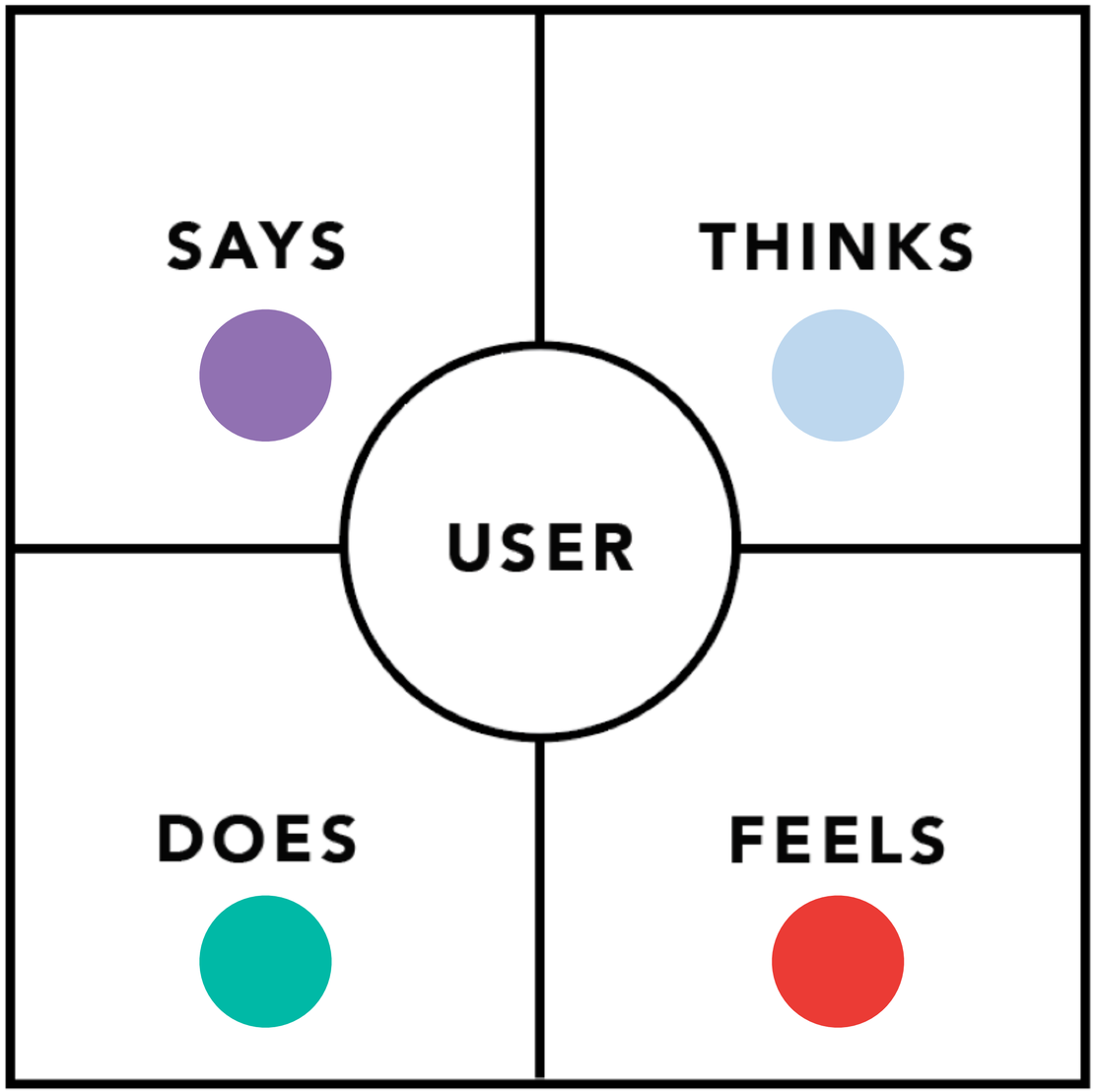

Img: https://www.nngroup.com/articles/empathy-mapping/

“Is my profile public?”

Refreshes page several times.

The signup process is confusing.

confused: too many contradictory prices



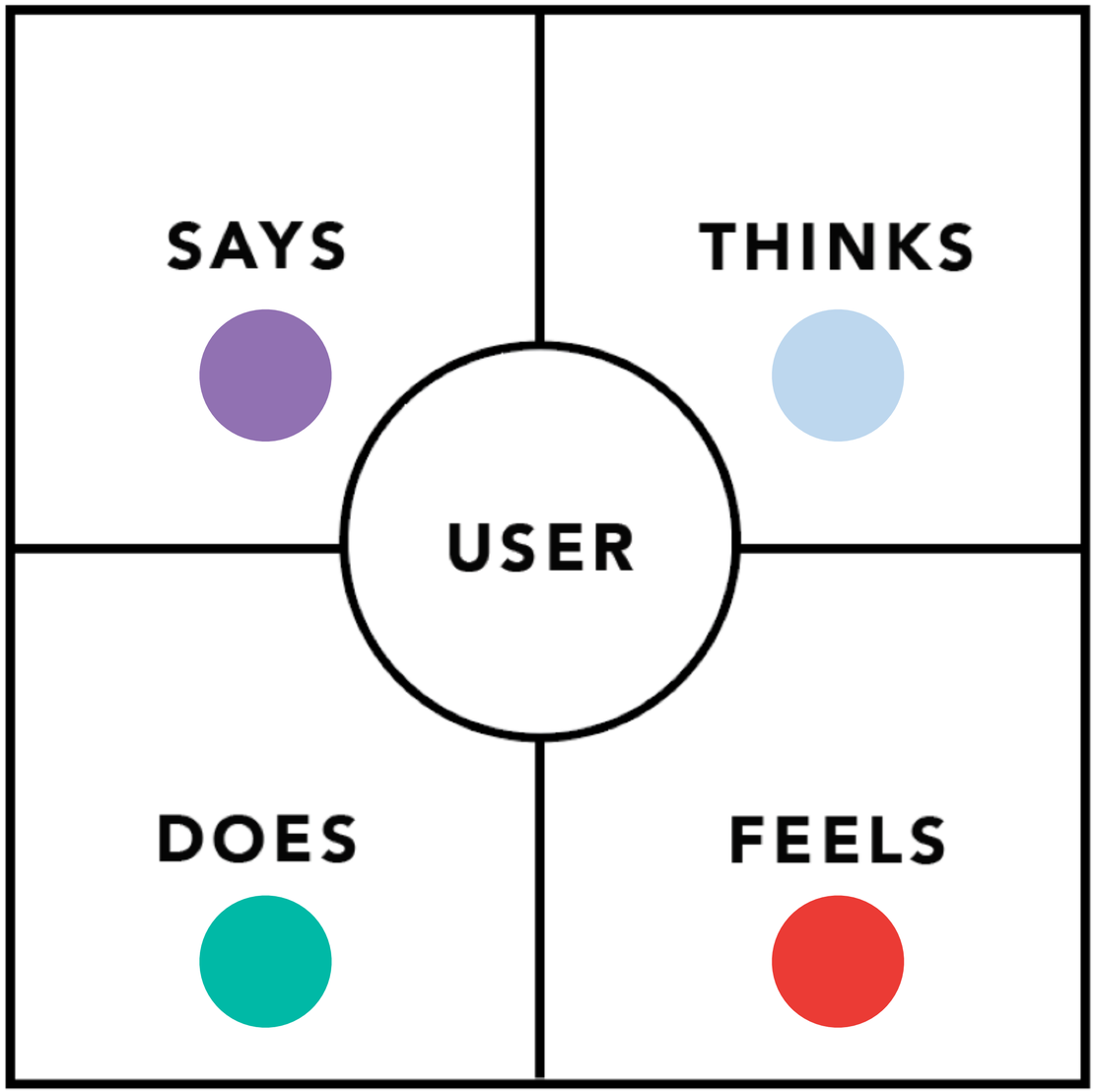

Img: https://www.nngroup.com/articles/empathy-mapping/, https://assets.interaction-design.org/literature/topics/empathy-mapping

Direct quotes and expressed opinions
e.g. interview, survey

Observed actions and behaviours
e.g. usability test

Research-informed thoughts, beliefs
eg. interview, think-aloud

Research-informed emotion, frustration
e.g. interview, observation




Img: https://www.nngroup.com/articles/empathy-mapping/, https://assets.interaction-design.org/literature/topics/empathy-mapping

# User story

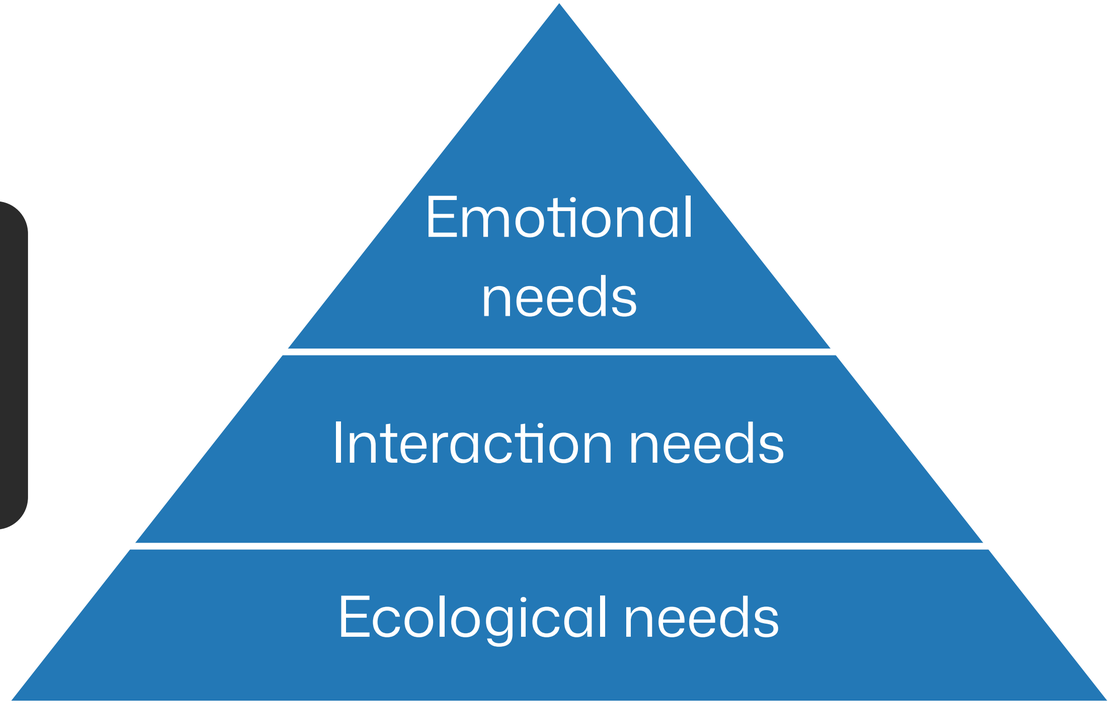
## gather user wants and needs as inputs

Who

What

„As a **dispatcher**, I want to **see a list** of all vehicles and their current locations, so that I can **monitor ongoing deliveries.**“

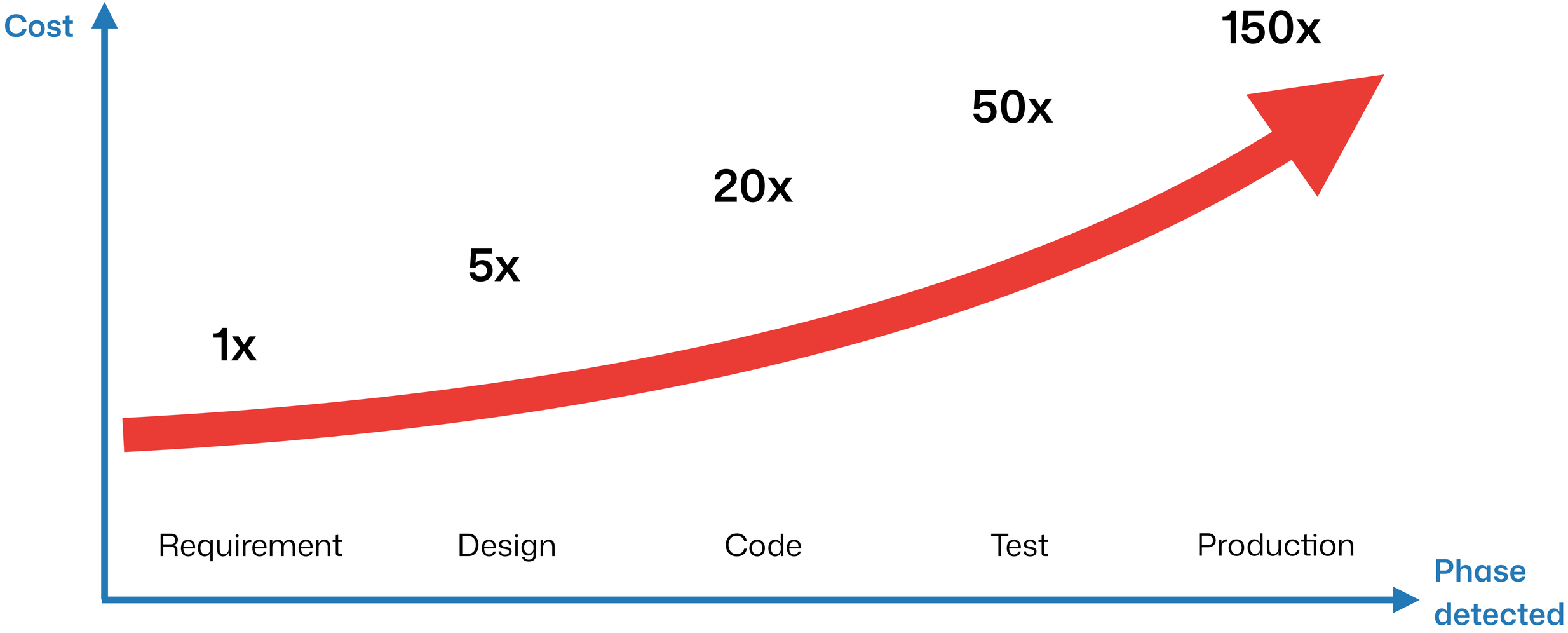
Why



# Prototyping

Which is more 🤑?
Discard a bad idea in one minute or program it 3 months?

# Cost of change curve

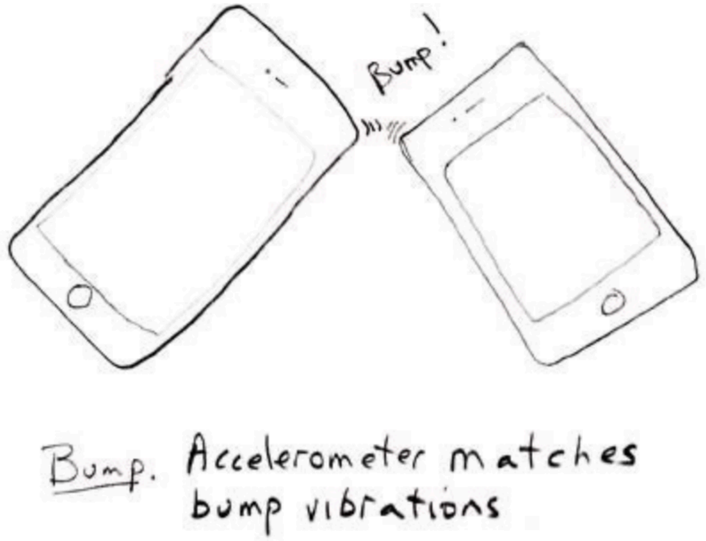


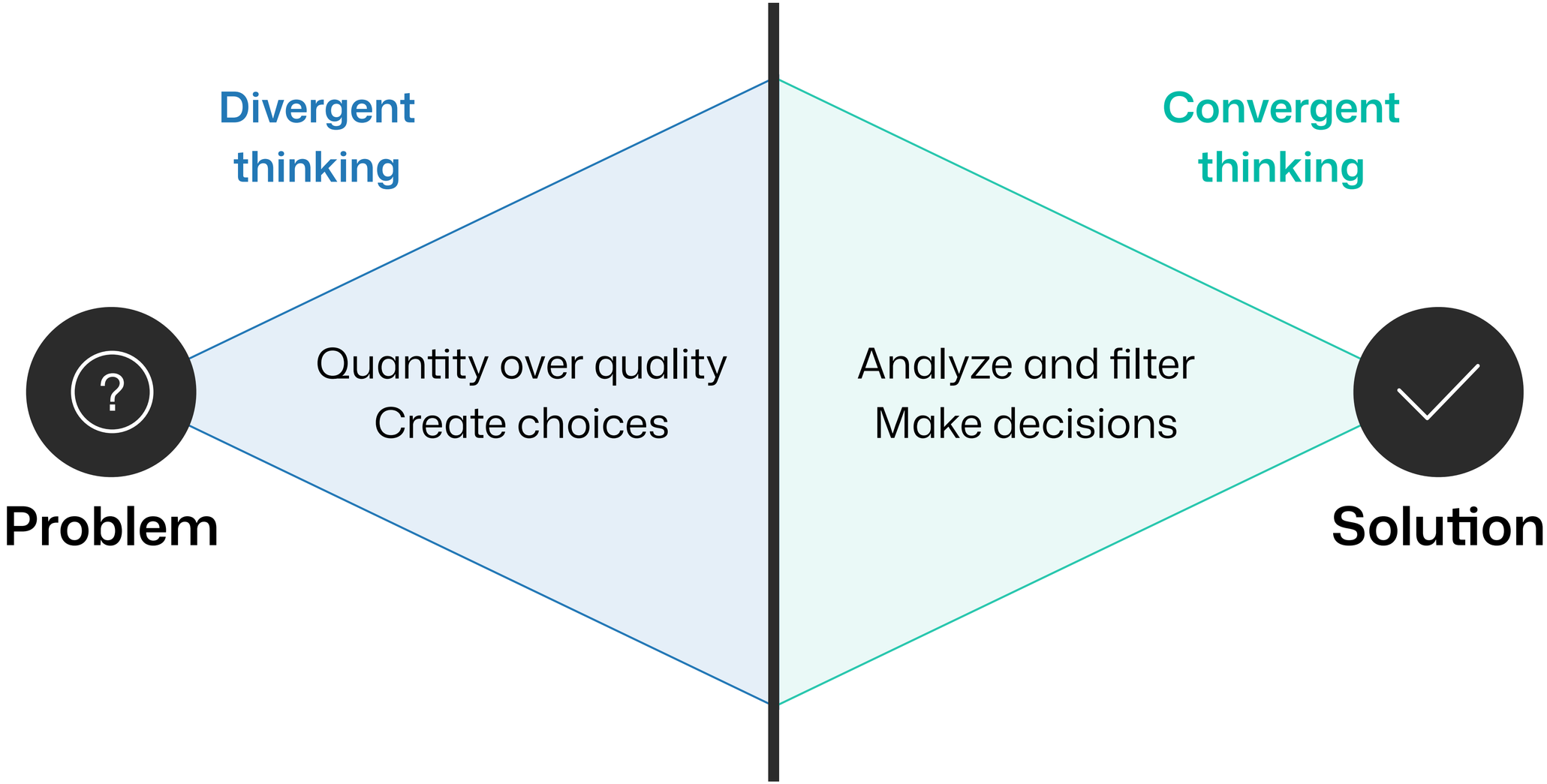
# If you’re not sketching, you’re not doing design

**Exploration** • Design is a process of creation and exploration, and sketching a visual medium for exploration of multiple design alternatives

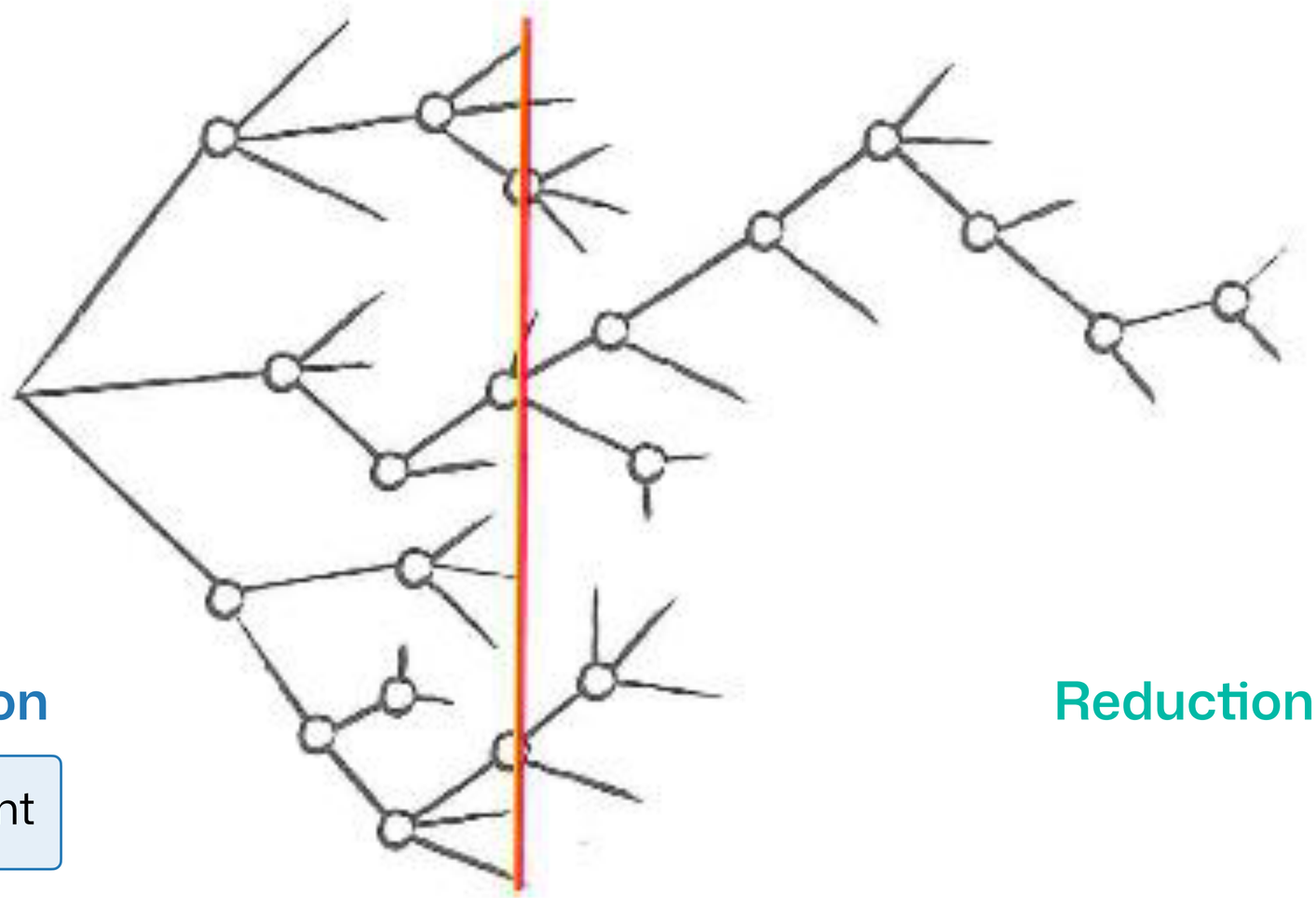
**Communication** • Sketching gives ideas a visible & tangible form

**Thinking tool** • Sketching supports creative thinking and iterative problem-solving




Divergent thinking
Convergent thinking
Quantity over quality
Create choices
Analyze and filter
Make decisions
Problem
Solution

# Develop many ideas



**Elaboration**

no judgment

**Reduction**

# 👋 Hands on „Crazy Sketching“

## Use sketching as a thinking tool

- Divide a sheet / Figma frame into 8 sections, then sketch 8 different ideas **individually**
- Quality doesn’t matter → focus on **quantity** and exploring many solutions

Afterwards, use dot voting (everyone has 🟢🟢🟢🟢) to vote the ideas:

✓ Which idea is genuinely new?

✓ Which idea actually solves the user’s problem?

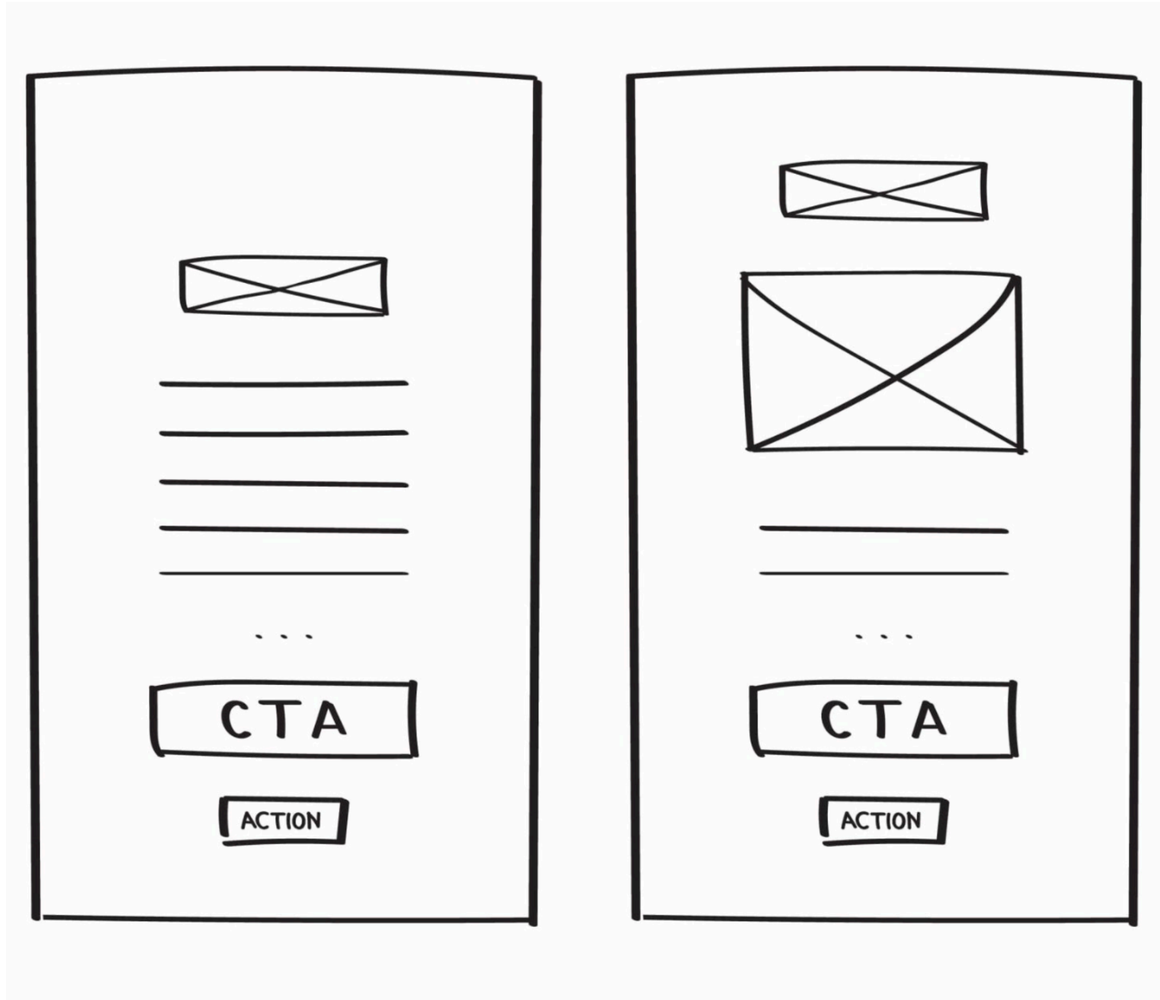
✓ Which idea is easy to implement?

Show your top idea in plenum

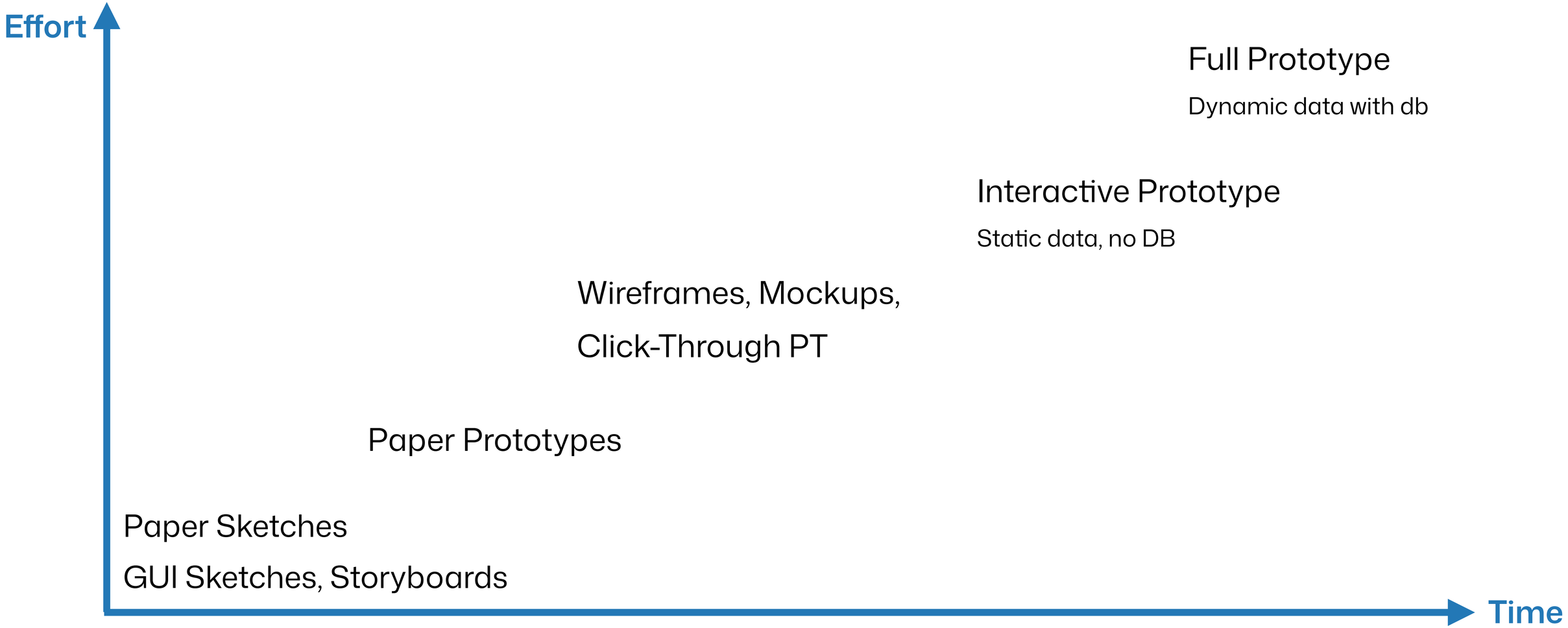
# When you have an idea 💡 would you start with the implementation right away?

# Why prototyping?

- Make „quick & cheap“ errors
- Find errors early
- Communicate feedback
- Explore alternatives

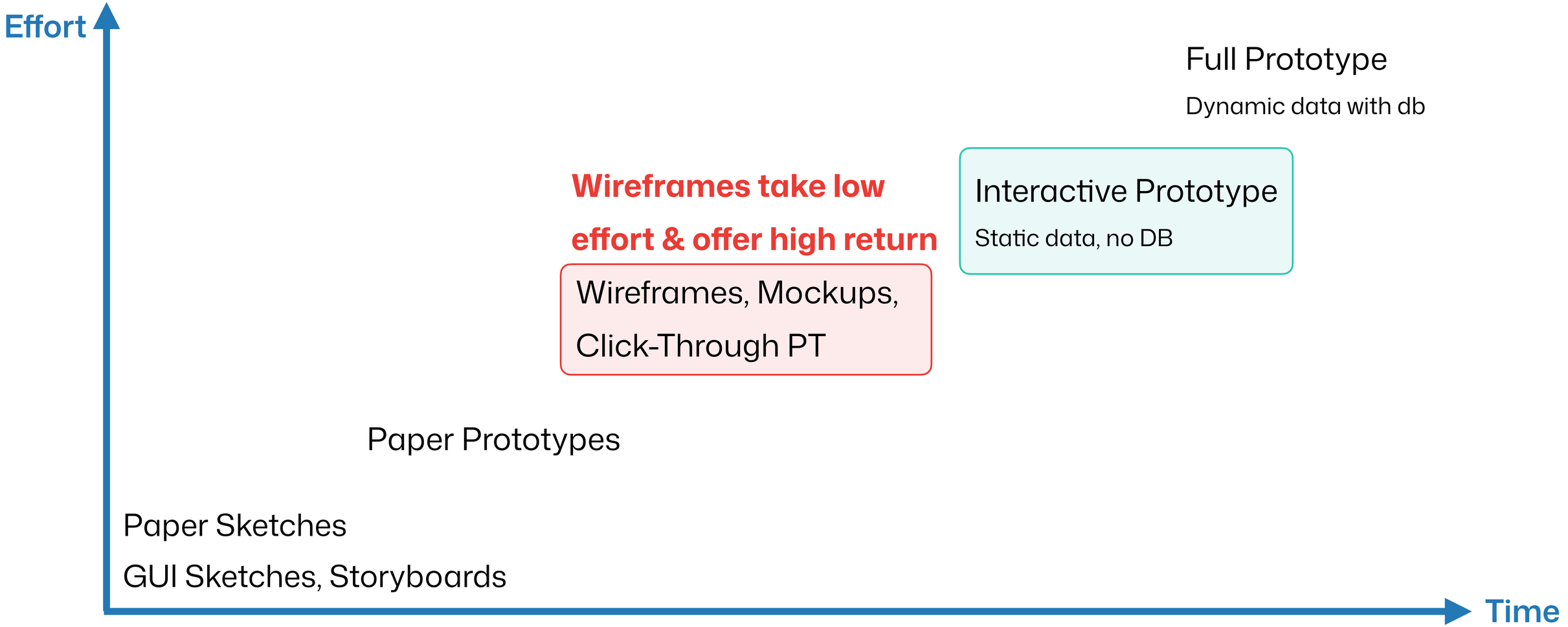


# Why prototyping?



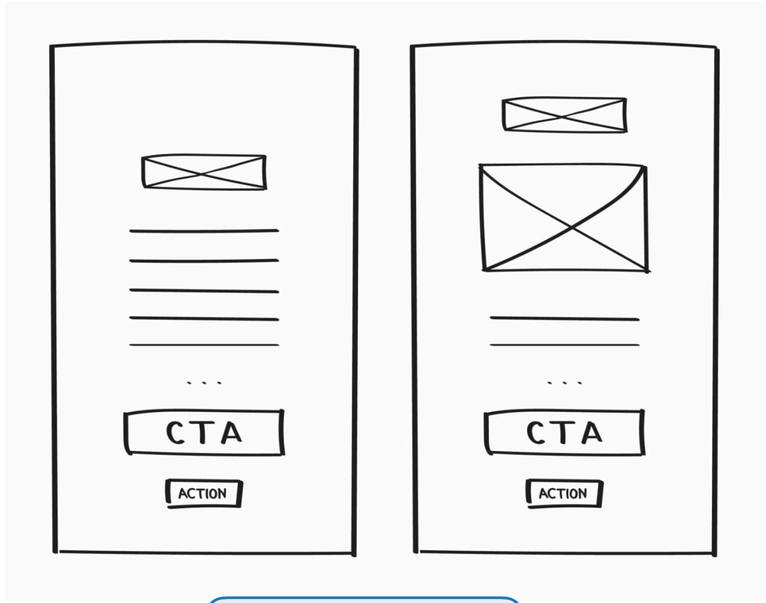
# Why prototyping?

Use appropriate tools, don't waste time by coding too early!

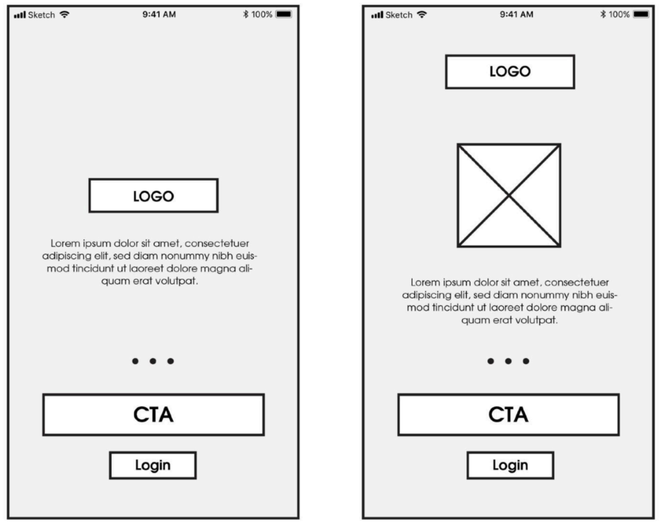


# Wireframe • static

Visual template of screen design, communicating the **idea** and **structure** of the page



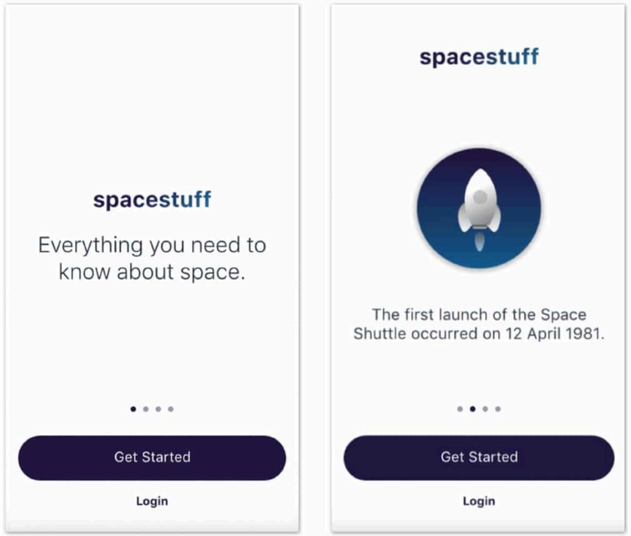

Low fidelity




Medium fidelity

# Prototype • interactive

Give look at **interaction, style** & **positioning** of UI elements




High fidelity

Source: Tsvetelina Lazarova https://mentormate.com/blog/low-fidelity-wireframes-vs-high-fidelity-wireframes/

## Low fidelity

- Simple
- Low-cost
- Quick & easy to change

🔄 **Low-level feedback** e.g. idea, concept, structure, wording

## High fidelity

- More detailed
- Needs more time
- Closer to final product

🔄 **High-level feedback** e.g. style, color, icon

# Button up ⬆️

## Starts with existing work practice

**Simplify existing transfer form**

- ✓ **Biases & constraints** • existing user preferences & established practices influence decisions
- ✓ Less likely leading to innovative possibilities
- ✓ Bottom-up design fills a **niche**

# Top down ⬇️

## Rethinks abstract work activity

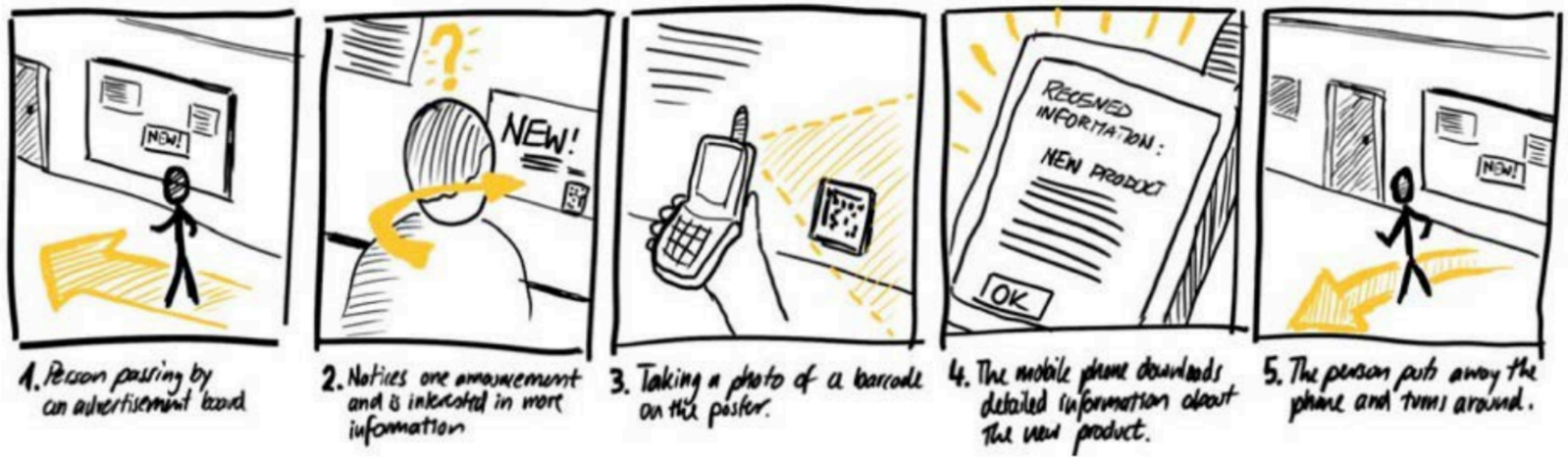
**Explore new money transfer way**

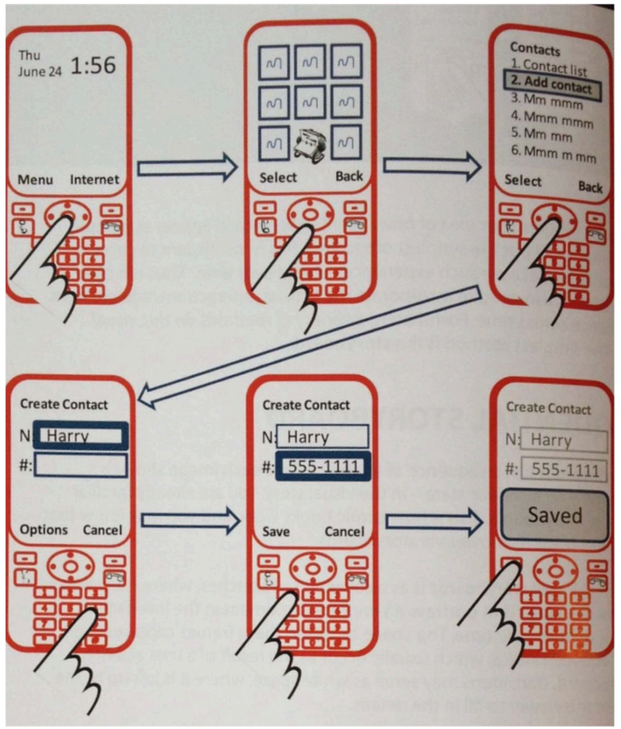
- ✓ **Biases & constraints** • Rethinks how work could be accomplished
- ✓ Questions inherited assumption
- ✓ Top-down design can **create a niche**

# Storyboard

- Sketches show a **moment** in a system, storylines add the **time** component
- Show how persona uses the system to achieve goals
- Produces empathy and shows context when system is used
- **Created from scenario** by dividing it into several steps
- Emphasise interactions and motions!

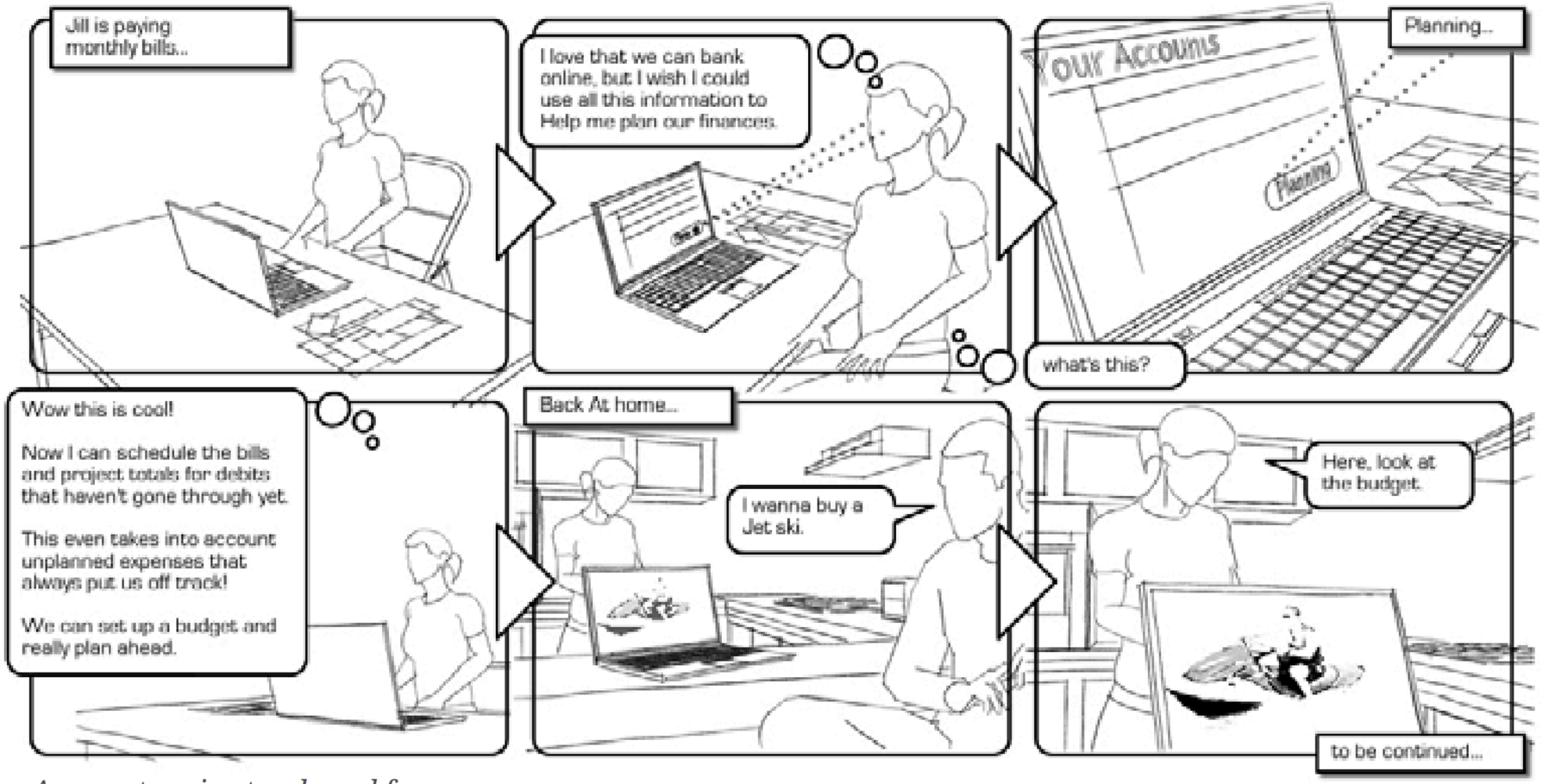
# Storyboard


NEW!
NEW!
RECEIVED INFORMATION:
NEW PRODUCT
OK
NEW!
1. Person passing by an advertisement board
2. Notices one announcement and is interested in more information
3. Taking a photo of a barcode on the poster.
4. The mobile phone downloads detailed information about the new product.
5. The person puts away the phone and turns around.


Thu
June 24 1:56
Menu Internet
Select Back
Contacts
1. Contact list
2. Add contact
3. Mm mmm
4. Mmm mmm
5. Mm mm
6. Mmm m mm
Select Back
Create Contact
N: Harry
#:
Options Cancel
Create Contact
N: Harry
#: 555-1111
Save Cancel
Create Contact
N: Harry
#: 555-1111
Saved

# Storyboard

convey understanding of users **interests and experiences** to stakeholders in engaging, emotionally appealing way

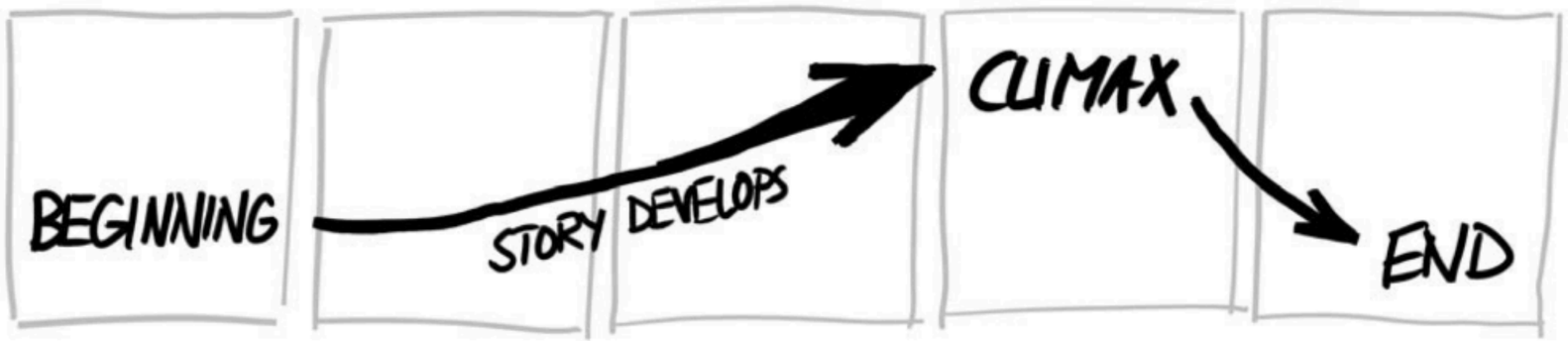



*A user story in storyboard form*

[Source: Anderson, 2010]

# Storyboard

Develop a storyline




conclusion

Where does the interaction take place?

What is the problem?

What is the task a person is trying to do?

Which people are present / What are their actions?

What objects/devices do they use?

What are the input/outputs?

How do their actions solve the problem?

# Storyboard

## Camera shots



**Extreme long shot (wide shot)**
A view showing details of the setting, location, etc.



**Long shot**
Showing the full height of a person



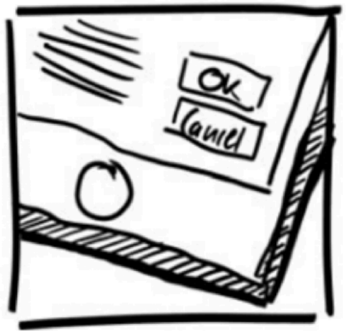
**Medium shot**
Show's a person's head and shoulders



**Over-the-shoulder shot**
Looking over the shoulder of a person



**Point of view shot (POV)**
Seeing everything that a person sees themselves




**Close-up**
such as showing details of a user interface a device the person is holding

# New team skills

## Teams need to understand & evaluate AI & write „effective“ prompts

What is still needed:

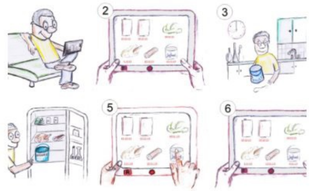
- ✓ Design skills
- ✓ Domain knowledge
- ✓ Creativity



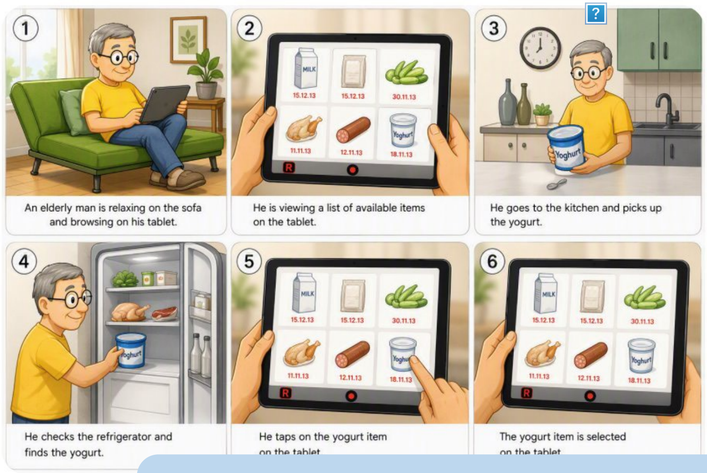
👉 AI doesn't replace design,
but it changes the way we work

# Generate storyboard from sketch

Copilot (GPT-5)



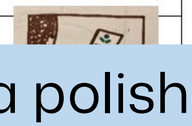
Create a polished storyboard based on the hand-drawn one i uploaded.

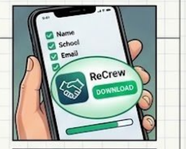



„Create a polished storyboard based on the hand-drawn one I uploaded.“

# Generate storyboard from sketch

Gemini Flash 3.6

| Key-frame | Beschreibung | Sketch |
|---|---|---|
| 1 | Sebastian steht gemeinsam mit seinem Freund Lukas an der Bushaltestelle. Wie jeden Tag pendelt er von Linz zur FH Hagenberg. Mit einem Seufzer meint er: „Ganz ehrlich, Studieren ist inzwischen richtig teuer geworden. Ich weiß gar nicht mehr, wie ich mir meine Wohnung und so weiterhin leisten soll …“ |  |
| 2 | Lukas nickt verständnisvoll. „Ja, ich versteh dich total!“, sagt er mitfühlend. Sebastian blickt nachdenklich dem einfahrenden Bus entgegen. „Weißt du, was ich am schwierigsten finde?“, fragt er schließlich. „Diese ganzen Jobportale – die sind alle so unübersichtlich und völlig unpersönlich! Die meisten Unternehmen suchen ohnehin nur nach Personen mit Berufserfahrung, wir Studenten werden da einfach ignoriert. Und das, obwohl wir Studierende der FH eigentlich sehr praktisch ausgebildet werden.“ |  |
| 3 | Die beiden steigen in den Bus ein und setzen sich auf einen freien Zweierplatz. Da fällt Lukas etwas ein: „Wusstest du, dass es eine App gibt, die genau auf solche Bedürfnisse eingeht? Sie heißt *ReCrew*. Das ist ein oberösterreichweites Jobportal speziell für Studierende der FH OÖ. Unternehmen können dort Nebenjobs und Praktika ausschreiben.“ |  |
| 4 | Überrascht holt Sebastian sein Handy heraus und sucht im App-Store nach *ReCrew*. „Die Registrierung ist super einfach“, erklärt Lukas. „Du musst nur kurz deine persönlichen Daten eingeben – und schon bist du dabei.“ |   |
| 5 | Lukas behält Recht: Die Registrierung funktioniert schnell und unkompliziert. Sebastian ist begeistert – er ho… keine Sorgen mehr um seine finanzie… |  |

| Key-frame | Beschreibung | Sketch |
|---|---|---|
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| 5 | **Lukas-Registration:** Die Registrierung funktioniert schnell und unkompli… …ch |  |

„Create a polished storyboard based on the hand-drawn one I uploaded.“

# 👋 Hands on

## Create (1) a context scenario and then (2) a storyboard with AI

Start with your **requirement** from last lesson → then create a storyboard

Discuss in plenum

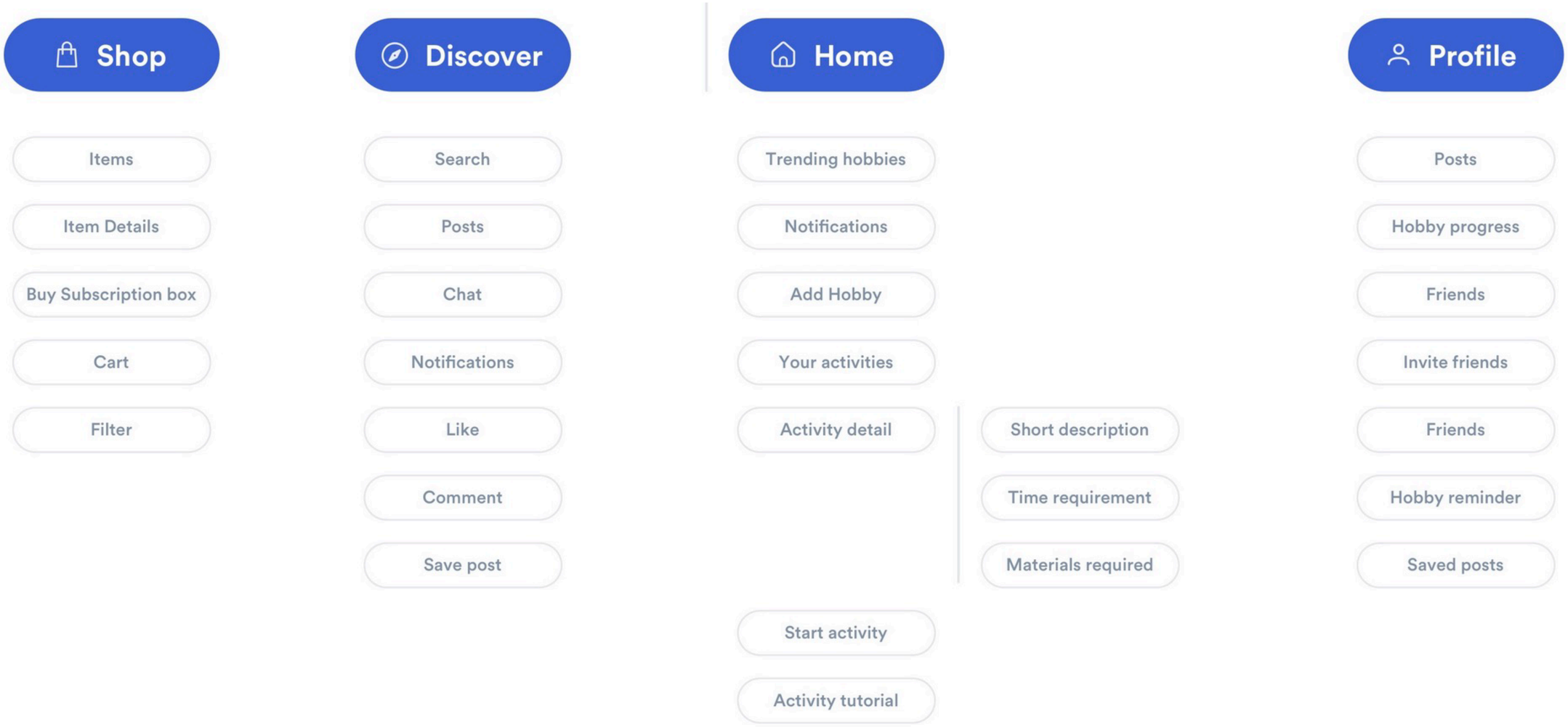
# Information Architecture

## Structures a product's content and functions

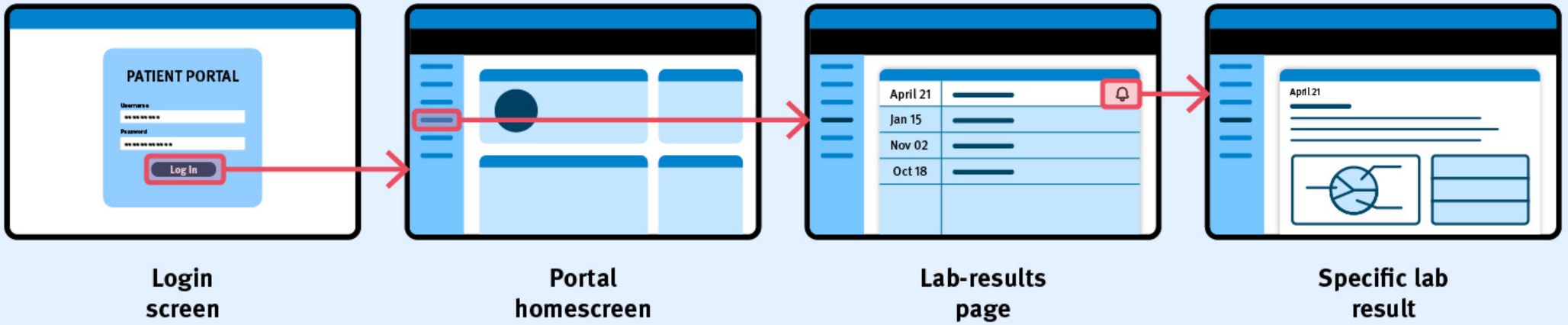
## How is information organised and found?

- ✓ It organizes, structures, and labels content so users can find what they need
- ✓ Think of it as the full plan of all the information on a website

# Information Architecture



# User Flow




**Wireflows** document the ideal or typical **pathway** through screens that need to be visited in order to accomplish a given **task**

[https://www.nngroup.com/articles/user-journeys-vs-user-flows/]

# 👋 Hands on „IA“

## Describe the Information Architecture of your prototype



# 👋 Hands on „AI Wireframe“

## First Requirement → creation of AI-Wireframe

- ✓ What assumptions did the AI make?
- ✓ What was not defined in the requirements?
- ✓ Turn a structured idea into a prototype
- ✓ Discuss possible solutions

**A wireframe shows the idea and structure of the page, not the design!**

- ✓ Greysacle, Rectangles, Lines, Text, easy Icons
- ✓ No fancy colors, no UI-details



# References


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